

DSA STUDY BLUEPRINT SERIES

27. Maths for DSA One Shot

Mathematical Primes, Greatest Common Divisors, and Primes Finder

Section 04 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Sieve of Eratosthenes:** Tracks primes by marking out composite multiples.
- **GCD:** Euclidean division algorithm: $\text{gcd}(a, b) = \text{gcd}(b, a \% b)$.

Memory & Execution Blueprint

Sieve Marking (N = 10):

Primes = { 2, 3, 5, 7 } • Mark multiples: 4, 6, 8, 9, 10
compiled as composites.

C++ Implementation Reference

Standard code layout and syntax

```
int gcd(int a, int b) {  
    return b == 0 ? a : gcd(b, a % b);  
}  
// Sieve: vector prime(N, true); mark multiples up to sqrt(N)
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Sieve of Eratosthenes	$O(N \log(\log N))$ time, $O(N)$ space
GCD Euclidean	$O(\log(\min(a,b)))$

Key Practices & Gotchas

- **Important:** Outer loop of Sieve can terminate safely at `sqrt(N)`.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace Sieve of Eratosthenes index markings up to limit $N = 10$:

Active Prime (p)	Multiples to Mark Composite	Primes Set State	Status
2	4, 6, 8, 10	{2, 3, 5, 7}	Multiples marked as composite
3	9	{2, 3, 5, 7}	Multiples marked as composite

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Segmented Sieve:** Optimizes memory footprints for very large limits by processing sub-ranges individually in memory arrays.
- **Euclidean GCD:** Fast division steps checking divisibility remainders recursively.
- **Related LeetCode Problems:** #204 Count Primes, #1979 Find Greatest Common Divisor of Array.